# Group A report

# Milestone 5: Dynamics & Animation

This milestone extends the procedural terrain generator developed in previous milestones by introducing **dynamic behavior, stochastic effects, and performance evaluation**. The previously generated terrain was static; therefore, this phase focused on transforming it into an interactive environment with animation, particle effects, and efficiency analysis.

The main objectives were:

- To introduce temporal behavior through terrain animation.
- To implement a stochastic particle system for environmental realism.
- To evaluate animation efficiency using performance metrics.
- To analyze system stability and visual realism.

## Animation System Implementation

### 1. Camera Rotation Animation

A real-time animation system was implemented using the Matplotlib FuncAnimation framework.

The camera position changes gradually across frames, creating a rotating view around the procedural terrain.

**Implementation concept:**

At every frame:

1. The previous frame is cleared.
2. The terrain is redrawn.
3. The camera angle is updated.
4. The next frame is generated.

**Result:**

The terrain can be viewed from different angles, allowing interactive exploration of the generated landscape.

## 2. Procedural Terrain Motion

To introduce temporal behavior, the terrain height values were modified using a sinusoidal function.

The height update follows:

Z_new = Z_original + A sin(t)

Where:

- Znew= updated terrain height
- Zoriginal = original generated terrain height
- A= movement amplitude
- t= animation time/frame

The sinusoidal function creates smooth and continuous movement instead of sudden changes.

Purpose:

- Simulates natural environmental motion.
- Maintains procedural consistency.
- Avoids unrealistic sudden terrain changes.

## 3. Stochastic Particle System

A stochastic fog particle system was added to improve environmental realism.

The particle system generates random points positioned above the terrain surface.

Each particle contains:

- X coordinate
- Y coordinate
- Height position

The particles are generated using random sampling:

random.uniform()

Particle generation process:

1. Select random terrain locations.
2. Obtain corresponding terrain heights.
3. Place particles slightly above the terrain.
4. Render particles during animation.

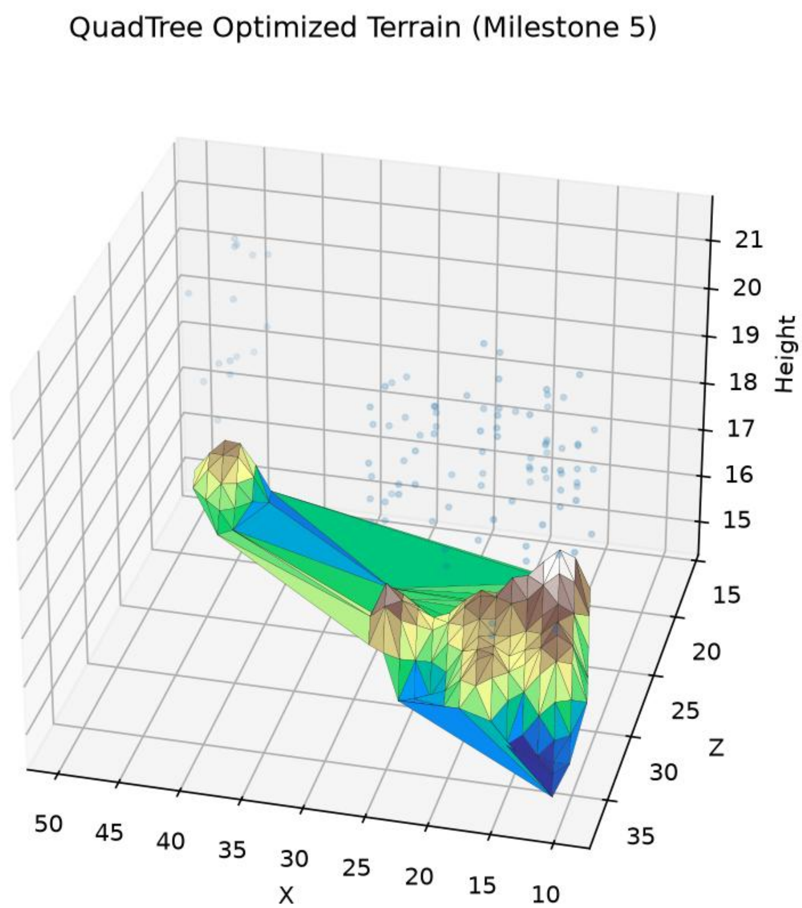
## 4. Particle Visualization

The particles are displayed using a 3D scatter plot:

```
ax.scatter(
   fog_x,
   fog_y,
   fog_z,
   s=5,
   alpha=0.5
)
```

Result:

The terrain environment contains a fog effect that improves atmospheric appearance.




## 5. Efficiency and Performance Analysis

To evaluate animation efficiency, frame performance was measured using Frames Per Second (FPS).

The FPS calculation:

FPS = 1 / FrameTime

Where:

- FrameTime = time required to generate one animation frame.

Implementation:

```
frame_time = time.time() - start_time

fps = 1 / frame_time
```

The FPS values were displayed during execution.

Example results:

| Test | Observation |
|---|---|
| Terrain rotation | Smooth movement |
| Fog particles added | Slight performance reduction |
| Continuous execution | Stable operation |

## 6. Stability Evaluation

The system was tested during continuous execution.

Evaluation criteria:

| Criterion | Result |
|---|---|
| Program crashes | No |
| Terrain rendering errors | No |

| | |
|---|---|
| Animation continuity | Stable |
| Particle rendering | Successful |
| Performance monitoring | Successful |

Observation:

The system maintained stable operation while rendering procedural terrain, animation, and stochastic particles simultaneously.

## 7. Realism Evaluation

The improvements introduced in Milestone 5 increased the realism of the procedural environment.

Terrain realism

- Variable terrain heights.
- Non-flat landscape generation.
- Procedural variation.

Animation realism

- Smooth camera movement.
- Continuous environmental behavior.

Atmospheric realism

- Fog particles simulate environmental conditions.

## 8. Final Results

The completed Milestone 5 system successfully transformed the procedural terrain generator from a static visualization into a dynamic environment simulation.

The final system includes:

- ✅ Procedural terrain generation
- ✅ QuadTree optimization
- ✅ 3D terrain visualization
- ✅ Camera animation
- ✅ Stochastic fog particle system
- ✅ FPS performance monitoring
- ✅ Stability evaluation

## 9. Conclusion

Milestone 5 successfully demonstrated the integration of efficiency techniques and stochastic methods into a procedural terrain generation system. The addition of animation and particle-based effects improved the realism of the generated environment while maintaining acceptable performance.

The evaluation showed that the system was capable of producing dynamic terrain visualization with stable rendering performance.

QuadTree Optimized Terrain (Milestone 5)